PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer | NLP & Document AI

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 5+ years of experience across NLP, data science and research, including deploying machine learning systems to production. Specialised in document classification and named entity recognition for Croatian administrative text. Experienced in taking models from dataset preparation and experimentation through model serving and production monitoring, using Python, MLflow and Azure. Currently pursuing a PhD in computer science focused on Croatian-language NLP.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | Slavonica Document AI d.o.o. | Osijek

03/2023–Present

- Developed and deployed a classifier for scanned incoming public-administration documents, achieving 94% accuracy and 0.91 macro F1 across 23 classes; the service processes approximately 15,000 documents per day.
- Fine-tuned XLM-R and BERTić transformer models for document classification and named entity recognition in Croatian administrative text, including persons, organisations, OIB numbers, addresses and legal references.
- Built an annotated dataset of 4,000 documents and developed annotation guidelines; achieved an inter-annotator kappa of 0.87.
- Implemented model serving with FastAPI and Docker on Azure App Service; used Azure ML and MLflow for experimentation and model registry.
- Established weekly production monitoring for text length, vocabulary and predicted-class distribution, with alerts for PSI above 0.2. Detected a client-side OCR change associated with a 9-point accuracy drop and helped resolve it within a week.
- Mentor two student interns.

Data Scientist | Baranjska kreditna banka d.d. | Zagreb (remote from Osijek)

09/2021–02/2023

- Developed customer churn prediction models using gradient boosting and LightGBM for the sales team.
- Built a customer complaint topic-classification workflow for customer-service reporting, progressing from TF-IDF and logistic regression to fastText; classified 11 topics for weekly reporting to customer service management.
- Prepared SQL reports from an Oracle data warehouse.

Research Project Assistant | Faculty of Applied Informatics, University of Drava |

10/2020–08/2021

- Created a corpus of Croatian legal texts and annotated data; wrote supporting Python scripts.

SELECTED PROJECT

Retrieval-Augmented Question Answering Demo | Personal project *2025*

- Built a weekend prototype for question answering over Croatian laws using a sentence-embedding model, FAISS and an open-weight LLM accessed through an API.
- Prototype only; not used in production and checked manually without formal evaluation.

EDUCATION

PhD studies in Computer Science | University of Drava *2022–Present*

Research topic: Named entity recognition in Croatian legal texts | Expected completion: 2027

MSc in Computer Science (Mag. ing. comp.) | University of Drava *2020*

CERTIFICATIONS & TEACHING

Microsoft Certified: Azure AI Engineer Associate | 2024

Guest Lecturer, NLP Summer School | 2025 (90-minute lecture)

TECHNICAL SKILLS

Programming and data: Python, SQL, pandas, NumPy

Machine learning and NLP: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy, sentence-transformers, TF-IDF

Production ML and MLOps: MLflow, model registry, FastAPI model serving, Docker, Azure ML, Azure App Service, Azure Functions, Evidently, production monitoring, data-drift monitoring

Data and tools: Oracle SQL, PostgreSQL, FAISS, Label Studio, Git, Linux

LANGUAGES

Croatian: Native | English: C1 (regular English-language client meetings) | German: B2

PUBLICATION

Zlatarić-Novosel, P. et al. (2024). "Named Entity Recognition for Croatian Administrative Documents." Regional workshop on Slavic language processing.